

Riley Sinema

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EDUCATION

Brigham Young University

Provo, UT

Master of Science in Computer Science

Sept 2024–Apr 2026

- Master's project: compared fine-tuning, retrieval, and hybrid LLM architectures for Text-to-SQL over an undocumented ~8,000-table enterprise production schema, using training and evaluation sets built from 26,500+ queries written by production data engineers
- The hybrid system, a QLoRA-fine-tuned open-source model paired with example retrieval, achieved the best results across all metrics, quadrupling execution accuracy over the retrieval-only baseline (4.4% to 17.4%)

Bachelor of Science in Computer Science

Sept 2020–Apr 2024

- BYU Academic Scholarship; Dean's List 2023

EXPERIENCE

Enzy

May 2025–Present

AI/ML Engineer

Lehi, UT

- Shipped an agentic chat system adopted by over 50% of the customer base for daily analytics and reporting
- Built end-to-end evaluation suites for three deployed agent features (chat, insights, and automations) with 178 test cases and 30+ evaluators spanning correctness, adversarial safety, and cost; results visualized in LangSmith to version the agent and hill-climb on metrics
- Executed a large-scale data migration from legacy MySQL to a sharded MongoDB cluster, cutting compute cost by over 60% and accelerating analytical queries

Brigham Young University Department of Computer Science

Jan 2024–Apr 2024, Sept 2024–Apr 2025

Teaching Assistant for Database Design

Provo, UT

- Provided comprehensive instruction and support to 80+ undergraduate students in database design principles, SQL query optimization, and NoSQL technologies, resulting in an 85% pass rate for the course and consistently positive student feedback
- Created engaging database design assignments and hands-on NoSQL labs that increased average student participation and led to a 90% assignment completion rate

Marching Order

May 2024–Aug 2024

Software Engineer

Remote

- Implemented critical components of the refactored architecture using modern design patterns, successfully migrating 20+ legacy features while maintaining backward compatibility
- Established automated testing protocols for refactored components, achieving 90% test coverage and reducing regression bugs by 75%

PUBLICATIONS

- Spencer Young, **Riley Sinema**, Cole Edgren, Andrew Hall, Nathan Dong, Porter Jenkins. "Assessing the Probabilistic Fit of Neural Regressors via Conditional Congruence." *European Conference on Artificial Intelligence (ECAI)*, Main Track, Bologna, Italy, 2025. arXiv:2405.12412

TECHNICAL SKILLS

Python, LLMs, RAG, LangChain/LangGraph, LangSmith, Agent Evals, LoRA/QLoRA fine-tuning, MLOps, AWS, SQL, MySQL, PostgreSQL, NoSQL, Redis, MongoDB, PyTorch, NLP, Hugging Face, Docker, Linux, Bash, Git, Java

VOLUNTEER EXPERIENCE

The Church of Jesus Christ of Latter-day Saints

Aug 2018–Apr 2020

Full Time Missionary

Leyte, Philippines

- Taught local residents about Church doctrine and participated in community service projects, working full time in Filipino

ADDITIONAL

Interests: Film and landscape photography, hiking and camping throughout Utah, skateboarding, snowboarding